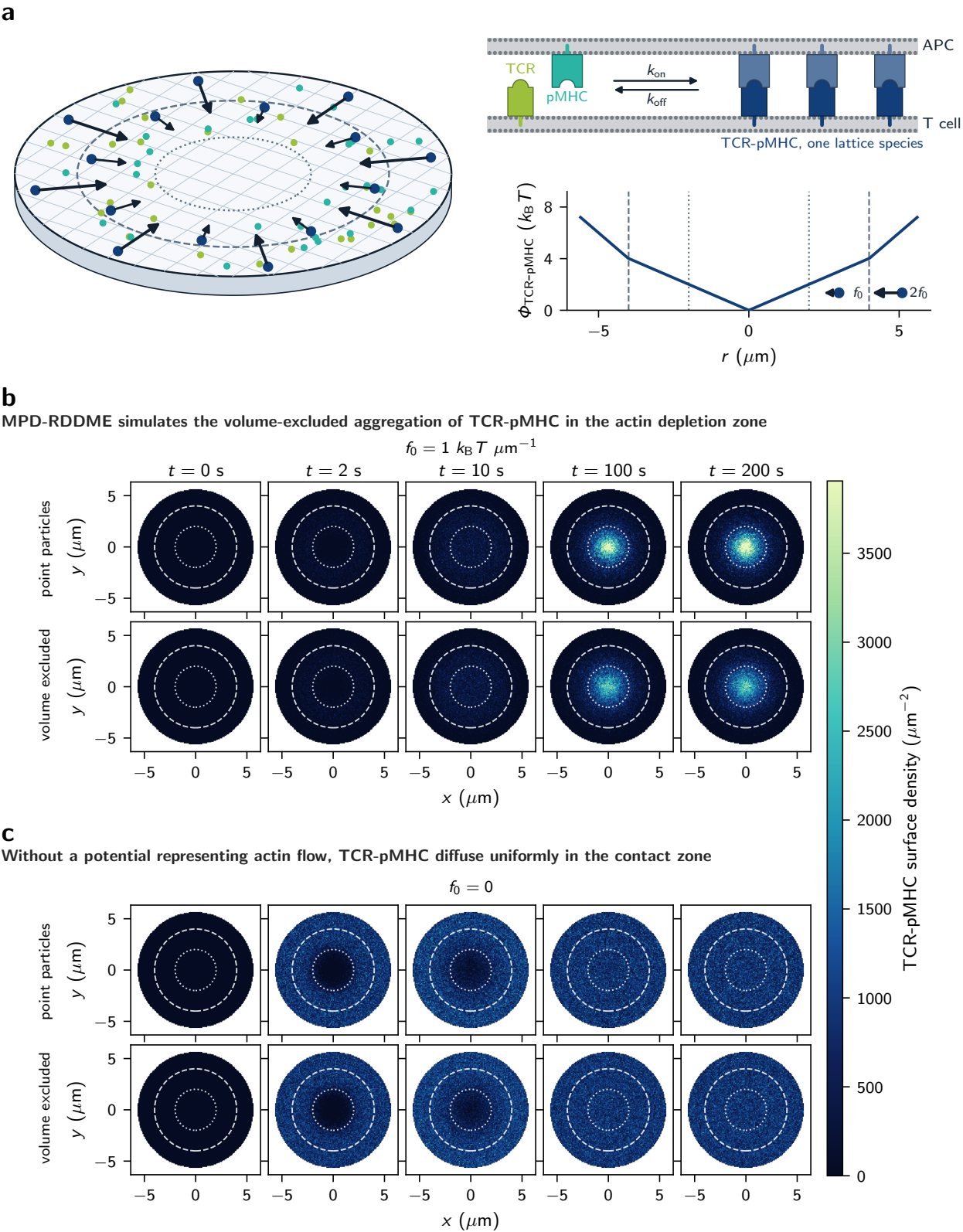

a
APC
TCR
pMHC
k_{on}
k_{off}
T cell
TCR-pMHC, one lattice species
$\phi_{\text{TCR-pMHC}}$ ($k_B T$)
8
4
0
f_0
$2f_0$
−5
0
5
r (μm)
b
MPD-RDDME simulates the volume-excluded aggregation of TCR-pMHC in the actin depletion zone
$f_0 = 1\ k_B T\ \mu\text{m}^{-1}$
$t = 0$ s
$t = 2$ s
$t = 10$ s
$t = 100$ s
$t = 200$ s
point particles
volume excluded
y (μm)
x (μm)
TCR-pMHC surface density (μm^{-2})
3500
3000
2500
2000
1500
1000
500
0
c
Without a potential representing actin flow, TCR-pMHC diffuse uniformly in the contact zone
$f_0 = 0$